na-tabpos

Sign, variation and **relative-position** tables for Typst

version 1.0.0 · built on vartable and CeTZ

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1 – Introduction

`na-tabpos` is a Typst package for Arabic (and any other) math documents. It is built on **vartable** and adds one thing: a **relative-position** row — curve against a line, or curve against curve — with the two arcs drawn at every intersection, like `\tkzTabPos` in LaTeX.

Everything else is exactly vartable: sign rows, variation rows, hatched undefined areas, `"||"` and `"0"` bars, added values, styles… The syntax of vartable is unchanged.

Main features

- The **difference row** `f(x) - g(x)` — a sign row placed right before a `"p"` row — is **1 cm** high by default (option `diff-height`).
- The words of the `"p"` row (above / below / crosses / at the point) **follow the language of the document** (`#set text(lang: ...)`): Arabic, French, English, Spanish, German, Italian, Portuguese, Dutch, Turkish, Russian. Any other language uses English. You can always give your own words.
- A height written in a `label` (for example `([title], 2cm, "s")`) is exactly that many centimetres.

2 – Installation and import

From the Typst Universe (when published):

```
#import "@preview/na-tabpos:1.0.0": *
```

Or copy the package folder next to your document and write:

```
#import "lib.typ": *
```

The package needs Typst ≥ 0.13 and loads CeTZ automatically. For Arabic text use an Arabic font (for instance **Amiri**) and `lang: "ar"`.

3 – Row types

Each element of `label` is an array `(title, type)` or `(title, height, type)`.

Type	Meaning
`"s"`	Sign row. The **difference row** `f(x) - g(x)` is just a sign row.
`"v"`	Variation row.
`"p"`	**Relative-position row** (new).

The function is called `tabpos` (the name `tabvar` is also exported and is the same function).

4 – The difference row and its height

The difference row is written as a normal sign row. When it is placed **immediately before** a `"p"` row, its height is `diff-height`, **1 cm** by default:

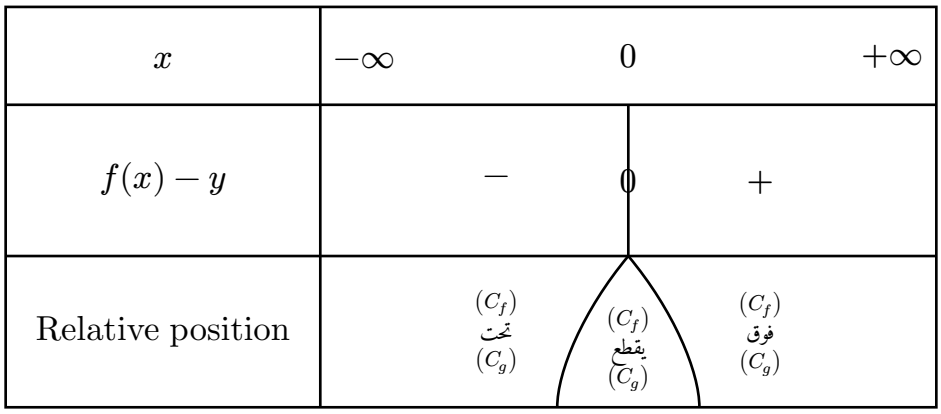
```
#tabpos(
  variable: $x$,
  label: (
    ([$f(x) - y$], "s"),     // difference row: 1 cm
    ([Relative position], "p"),
  ),
  domain: ($-oo$, $-1$, $1$, $+oo$),
  contents: (
    ($+$, ("0", $-$), ("0", $+$)),
    ("+", ($A(-1 ; 0)$, "-"), ($B(1 ; 0)$, "+")),
  ),
)
```

x	$-\infty$		-1		1		$+\infty$
$f(x)-y$		$+$	0	$-$	0	$+$	
Relative position		(C_f) فوق (C_g)	(C_f) يقطع (C_g) في النقطة $A(-1;0)$	(C_f) تحت (C_g)	(C_f) يقطع (C_g) في النقطة $B(1;0)$	(C_f) فوق (C_g)	

4.1 – Changing the heights

How	Effect
`diff-height: 1cm`	Default. Height of every difference row (a `"s"` row followed by a `"p"` row).
`diff-height: 2cm`	Another height for all difference rows of the table.
`diff-height: none`	Automatic height (the old behaviour of vartable, about 2.75 cm).
`([title], 1.5cm, "s")`	Height written in the row: it has priority over `diff-height`.
`pos-height: 3cm`	Default height of the `"p"` rows (3 cm).
`([title], 3cm, "p")`	Height of one particular `"p"` row.

```
#tabpos(
  variable: $x$,
  diff-height: 2cm,        // taller difference row
  pos-height: 2cm,         // shorter position row
  label: (([$f(x) - y$], "s"), ([Relative position], "p")),
  domain: ($-oo$, $0$, $+oo$),
  contents: (($-$, ("0", $+$)), ("-", ("0", "+"))),
)
```



5 – The `"p"` row

It is written exactly like a sign row, with these differences:

Entry	Result
`"+"`	**above**: the text `(C_f) / above / (C_g)`.
`"-"`	**below**: the text `(C_f) / below / (C_g)`.
`("0", "+")`	The two arcs at the left end of the interval, then **above**.
`(content, "-")`	The two arcs with the content (for example the point `$A(-1 ; 0)$`) under the text, then **below**.
`"|h|"`, `"h"`, `"h|"`, `"|h"`, `("||", "+")`…	Same as vartable: hatched areas and bars.

Use the strings `"+"` and `"-"` (not `$+$`) in a `"p"` row.

6 – Options of `tabpos`

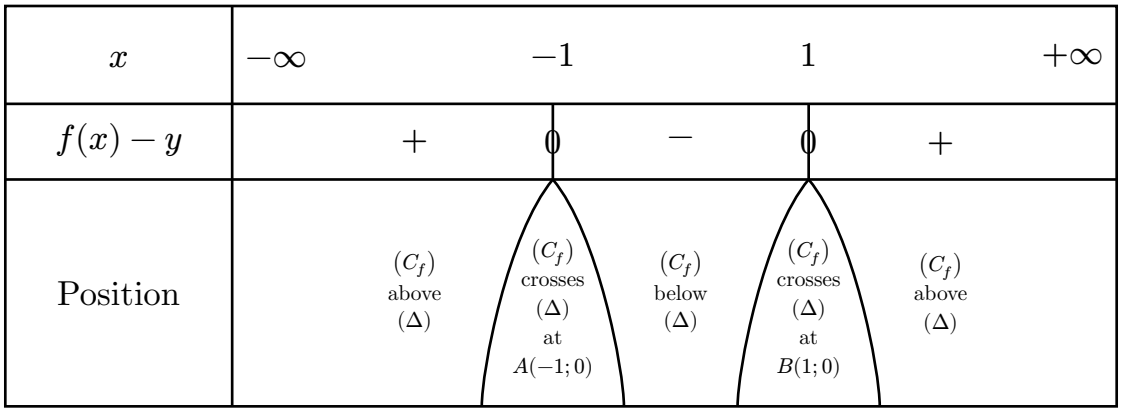
Option	Default	Meaning
`diff-height`	`1cm`	Height of the difference row (`"s"` row right before a `"p"` row); `none` = automatic.
`pos-height`	`3cm`	Default height of a `"p"` row, or per row: `([title], 3cm, "p")`.
`first`	`$(C_f)$`	First curve.
`second`	`$(C_g)$`	Second curve or the line, e.g. `$(Delta)$`.
`above`, `below`	`auto`	Words for the two positions. `auto` = word in the language of the document.

`cross`	`auto`	Word between the two curves at an intersection.
`at-point`	`auto`	Word before the point note.
`arc-width`	`0.94`	Half-distance between the two arcs (CeTZ units, 1 = 1 cm).

All the options of vartable (`hatching-style`, `values`, `table-style`, `first-line-height`, …) still work.

Words in English:

```
#tabpos(
  variable: $x$,
  second: $(Delta)$,
  above: [above], below: [below], cross: [crosses], at-point: [at],
  label: (([$f(x) - y$], "s"), ([Position], "p")),
  domain: ($-oo$, $-1$, $1$, $+oo$),
  contents: (
    ($+$, ("0", $-$), ("0", $+$)),
    ("+", ($A(-1 ; 0)$, "-"), ($B(1 ; 0)$, "+")),
  ),
)
```



7 – Language of the words

The default words of the `"p"` row follow the language set with `#set text(lang: ...)` — exactly like any other text of the document:

lang	above	below	cross	at-point
`ar`	فوق	تحت	يقطع	في النقطة
`fr`	au-dessus	au-dessous	coupe	au point
`en`	above	below	crosses	at the point
`es`	encima	debajo	corta	en el punto
`de`	oberhalb	unterhalb	schneidet	im Punkt
`it`	sopra	sotto	interseca	nel punto
`pt`	acima	abaixo	corta	no ponto
`nl`	boven	onder	snijdt	in het punt
`tr`	üstünde	altında	keser	noktasında
`ru`	выше	ниже	пересекает	в точке

```
#set text(lang: "fr")      // => "au-dessus", "au-dessous", "coupe", "au point"
#tabpos(/* ... */)
```

For another language, or other wording, write the words yourself: `above: [oberhalb]`, `below: [unterhalb]`.

8 – More examples

8.1 – Two curves

```
#tabpos(
  variable: $x$,
  second: $(C_g)$,
  label: (([$f(x) - g(x)$], "s"), ([Position], "p")),
  domain: ($-oo$, $0$, $+oo$),
  contents: (($-$, ("0", $+$)), ("-", ("0", "+"))),
)
```

x	$-\infty$		0		$+\infty$
$f(x) - g(x)$		$-$	0	$+$	
Position		(C_f) تحت (C_g)	(C_f) يقطع (C_g)	(C_f) فوق (C_g)	

8.2 – Forbidden value, $D_f = \mathbb{R} \setminus \{1\}$

```
#tabpos(
  variable: $x$,
  label: (([$f(x) - y$], "s"), ([Position], "p")),
  domain: ($-oo$, $1$, $+oo$),
  contents: (($-$, ("||", $+$)), ("-", ("||", "+"))),
)
```

x	$-\infty$		1		$+\infty$
$f(x) - y$		$-$	‖	$+$	
Position		(C_f) تحت (C_g)	‖	(C_f) فوق (C_g)	

8.3 – Hatched gap, $D_f =]-\infty;-2[\cup]2;+\infty[$

```
#tabpos(
  variable: $x$,
  hatching-style: hatch,
  label: (([$f(x) - y$], "s"), ([Position], "p")),
  domain: ($-oo$, $-2$, $2$, $+oo$),
  contents: (($-$, "|h|", $+$), ("-", "|h|", "+")),
)
```

x	$-\infty$	-2	2	$+\infty$
$f(x)-y$	$-$		$+$	
Position	(C_f) تحت (C_g)		(C_f) فوق (C_g)	

8.4 – Signs, variations, difference and position in one table

```
#tabpos(
  variable: $x$,
  label: (
    ([$f'(x)$], "s"),
    ([$f(x)$], "v"),
    ([$f(x) - y$], "s"),
    ([Position], "p"),
  ),
  domain: ($-oo$, $0$, $+oo$),
  contents: (
    ($-$, ("||", $-$)),
    ((center, $0$), (bottom, top, "||", $-oo$, $+oo$), (center, $0$)),
    ($+$, ("||", $-$)),
    ("+", ("||", "-")),
  ),
)
```

x	$-\infty$	0	$+\infty$
$f'(x)$	$-$	‖	$-$
$f(x)$	$0 \searrow -\infty$	‖	$+\infty \searrow 0$
$f(x) - y$	$+$	‖	$-$
Position	(C_f) فوق (C_g)	‖	(C_f) تحت (C_g)

9 – Files of the package

`lib.typ`	The whole package in one file (code derived from vartable 0.2.5, plus the `"p"` row, `diff-height` and the language-aware words). Exports `tabpos`, `tabvar`, `hatch`, `grille`, …
`typst.toml`, `LICENSE`	Package manifest and licences.
`README.md`, `README.ar.md`	Short presentation (English / Arabic).
`documentationEN.typ` / `.pdf`	This documentation.

`documentationAR.typ` / `.pdf`	The same documentation in Arabic.
`examples/`	Ready-to-compile examples (`.typ`, `.pdf`).
`fichier_de_test/`	Test files: heights, `"p"` options, hatching.

10 – Troubleshooting

- **The words of a `"p"` row are not in my language:** they follow `#set text(lang: ...)`; or give `above`, `below`, `cross`, `at-point` yourself.
- **`+` or `-` do not give the words:** in a `"p"` row write the strings `"+"` and `"-"`, not `$+$`.
- **Arabic letters are disconnected or missing:** set an Arabic font, e.g. `#set text(font: "Amiri", lang: "ar")`.
- **I want the old automatic height of the difference row:** write `diff-height: none`.

11 – Credits and licence

`na-tabpos` is © Naam Mohamed (MIT). The code of `lib.typ` is derived from **vartable** © Le_Foucheur (MIT) — the `LICENSE` file carries both copyrights.